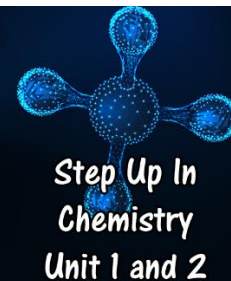


4.1 The Mole

Problems Worksheet



1. Determine the number of moles of:
 - a. Water in 301×10^{23} molecules of H_2O
 - b. Carbon dioxide in 602×10^{21} molecules of CO_2
 - c. Hydrogen chloride in 1,000,000 molecules of HCl
 - d. Nitrogen atoms in 1.806×10^{24} molecules of N_2 .
 - e. Atoms in 4.50×10^{23} molecules of CH_3COOH .

2. Determine the number of:
- a. Hydrogen peroxide molecules 1.55 moles of H_2O_2 .
 - b. Nitrogen dioxide molecules 0.100 moles of NO_2
 - c. Hydrogen chloride molecules 3.75 moles of HCl
 - d. Hydrogen atoms molecules 1.25 moles of NH_3 .
 - e. Atoms in 0.0675 moles of H_2SO_4 .

3. Determine the molar mass of the following:

a. Oxygen gas (O_2)

b. Silver chloride (AgCl)

c. Nitrous oxide (N_2O)

d. Chloroform (CHCl_3)

e. Ethanol ($\text{C}_2\text{H}_5\text{OH}$)

4. Determine the number of moles of:
- a. Iron atoms in 100.0 g of Fe

 - b. Ethanoic acid in 0.00632 g of CH_3COOH

 - c. Magnesium carbonate in 0.125 g of MgCO_3

 - d. Benzene in 245 kg of C_6H_6 .

 - e. Methanol in 35.9 g of CH_3OH